



FORT STREET HIGH SCHOOL

Name: \_\_\_\_\_

Teacher: \_\_\_\_\_

Class: \_\_\_\_\_

## 2012

### HIGHER SCHOOL CERTIFICATE COURSE

### ASSESSMENT TASK 3: TRIAL HSC

# Mathematics

**Time allowed: 3 hours**

(plus 5 minutes reading time)

| Outcomes Assessed                                                                                                                                                 | Questions |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Chooses and applies appropriate mathematical techniques in order to solve problems effectively                                                                    | 1-10      |
| Manipulates algebraic expressions to solve problems from topic areas such as geometry, co-ordinate geometry, quadratics, trigonometry, probability and logarithms | 11,13     |
| Demonstrates skills in the processes of differential and integral calculus and applies appropriate techniques to solve problems                                   | 12, 14    |
| Synthesises mathematical solutions to harder problems and communicates them in appropriate form                                                                   | 15, 16    |

## Total Marks 70

### Section I 10 marks

Multiple Choice, attempt all questions,

Allow about 15 minutes for this section

### Section II 90 Marks

Attempt Questions 11-16,

Allow about 2 hours 45 minutes for this section

## General Instructions:

- Questions 11-16 are to be started in a new booklet
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used

| Section I  | Total 10 | Marks |
|------------|----------|-------|
| Q1-Q10     |          |       |
| Section II | Total 90 | Marks |
| Q11        | /15      |       |
| Q12        | /15      |       |
| Q13        | /15      |       |
| Q14        | /15      |       |
| Q15        | /15      |       |
| Q16        | /15      |       |
|            | Percent  |       |

**SECTION I** Multiple Choice (10 marks)

**CIRCLE CORRECT RESPONSE**

1. What is the value of  $\frac{(1.49)^2 - 1.98}{\sqrt{11.62 + 8.34 \times 2.72}}$  correct to three significant figures?

A. 0.040

B. 0.0409

C. 0.041

D. 0.0410

2. Find the values that satisfy  $x^2 - x < 6$

A.  $x < -2, x > 3$

B.  $x < -3, x > 2$

C.  $-3 < x < 2$

D.  $-2 < x < 3$

3. If  $\log_a 5 = 1.03$  and  $\log_a 2 = 0.64$  then the value of  $\log_a 10$  is:

A. 2.06

B. 1.67

C. 0.6592

D. 3.2

4.  $\sum_{n=5}^{14} 17 - 2n =$

A. -11

B. -20

C. -18

D. -28

5. Expressed in radian measure,  $235^\circ$  is:

A.  $\frac{\pi}{235}$

B.  $\frac{235}{\pi}$

C.  $\frac{47\pi}{36}$

D.  $\frac{36\pi}{47}$

6. The third and seventh terms of a geometric series are 1.25 and 20 respectively. What is the first term?

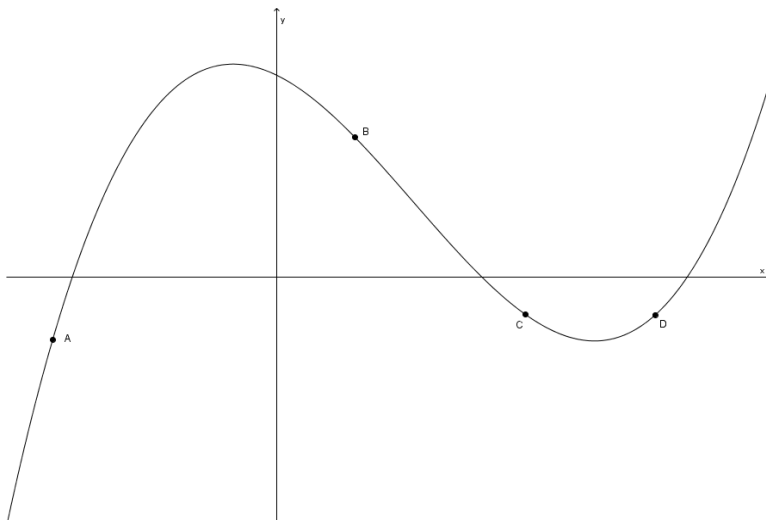
A.  $\pm 2$

B.  $\frac{5}{16}$

C.  $\pm 4$

D.  $\frac{5}{1024}$

7. State which point on the sketch fits the description  $y < 0, \frac{dy}{dx} > 0, \frac{d^2y}{dx^2} < 0$



A. A

B. B

C. C

D. D

8. What is the greatest value of the function  $y = 4 - 2 \cos x$ ?

- |    |   |    |   |
|----|---|----|---|
| A. | 2 | B. | 4 |
| C. | 6 | D. | 8 |

9. A game is played in which two coloured dice are thrown once. The six faces of the red die are numbered 3, 5, 7, 8, 9 and 11. The six faces of the white die are numbered 1, 2, 4, 6, 10 and 12. The player wins if the number on the white die is larger than the number on the red die. What is the probability that the player wins once in two successive games?

- |    |                  |
|----|------------------|
| A. | $\frac{7}{18}$   |
| B. | $\frac{11}{18}$  |
| C. | $\frac{77}{162}$ |
| D. | $\frac{77}{324}$ |

10. What is the centre and radius of the circle with the equation  $x^2 + y^2 + 6x - 8y - 11 = 0$ ?

- |    |                                 |
|----|---------------------------------|
| A. | Centre $(-3, -4)$ and radius 36 |
| B. | Centre $(-3, 4)$ and radius 36  |
| C. | Centre $(-3, -4)$ and radius 6  |
| D. | Centre $(-3, 4)$ and radius 6   |



**SECTION II** All necessary working must be shown

**Question 11** (15 marks) **Begin a NEW booklet**

**Marks**

(a) Evaluate  $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$  2

(b) If  $\sqrt{50} - 3\sqrt{75} + \sqrt{18} = a\sqrt{2} - b\sqrt{3}$ , find the value for  $a$  and  $b$ . 2

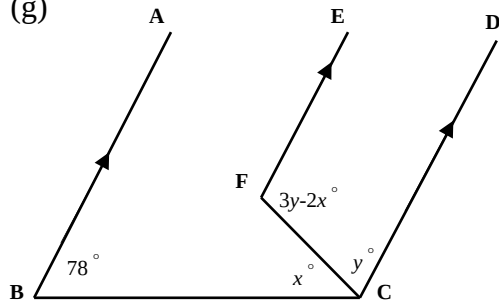
(c) State the domain and range of the function:  $f(x) = 2\sqrt{x-1} + 3$  2

(d) Evaluate  $a$ , in simplest exact form, if  $\int_1^3 \frac{a}{1+x} dx = \log_e 16$ . 2

(e) Solve for  $x$ :  $|2x - 1| = 3x - 4$  2

(f) For what values of  $k$  will  $x^2 - (k - 3)x + k = 0$  have equal roots? 2

(g) 3



$AB \parallel EF \parallel DC$ . Find the value of  $x$  and  $y$ , giving reasons.

**Question12 (15 marks) Begin a NEW booklet****Marks**(a) Differentiate with respect to  $x$ :

i)  $y = \sqrt{1 - x^2}$  1

ii)  $y = \frac{\cos x}{x}$  1

(b) Find:

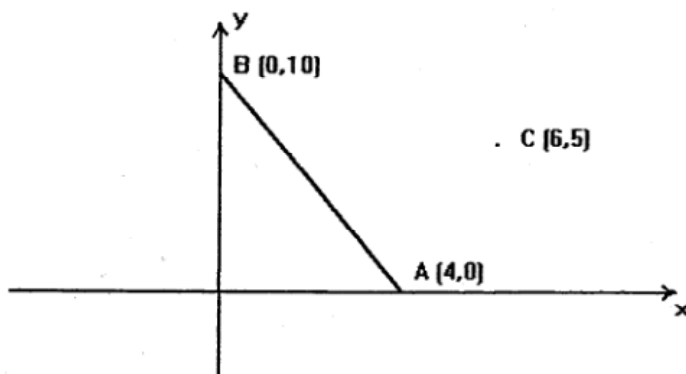
i)  $\int \frac{1}{e^{3x}} dx$  1

ii)  $\int (1 + 3\sec^2 \frac{x}{2}) dx$  1

(c) The gradient of a curve is given by  $f'(x) = 6x^2 - 6x + 5$ . If the curve passes through the point  $(2, 13)$ , find its equation. 2

(d) Find the equation of the normal to the curve  $y = \ln \left( \frac{2x-1}{x+1} \right)$ , at the point where  $x = 2$ . 3

(e) 6



The points A, B and C have coordinates  $(4, 0)$ ,  $(0, 10)$  and  $(6, 5)$  respectively as shown on the diagram. Copy the diagram into your booklets.

(i) Find the gradient of the line AB.

(ii) Show that the equation of the line through point C perpendicular to AB is given by

$$2x - 5y + 13 = 0. \text{ Label this line } q.$$

(iii) Find the perpendicular distance of B from the line CD, giving your answer in surd form with rational denominator.

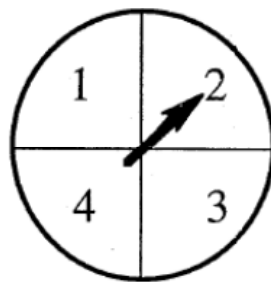
(iv) If the line  $q$  meets the  $x$ -axis at D, calculate the size of angle  $ADC$ , correct to the nearest degree.

**Question 13 (15 marks) Begin a NEW booklet****marks**

- (a) A party of walkers leave their base camp A at noon and walk in the direction  $150^\circ T$  until they reach a point B on the bank of the river. Unable to cross the river, they follow it on a bearing of  $015^\circ T$  and retire at 2 pm when they reach point C. The bearing and distance of C from A are  $120^\circ T$  and 2.5 km respectively.

**4**

- i) In your booklet, draw a diagram and mark on it the given information, in the correct positions.
- ii) Find the distances AB and BC.
- iii) Hence, calculate the average speed, correct to one decimal place, of these walkers from A to C.

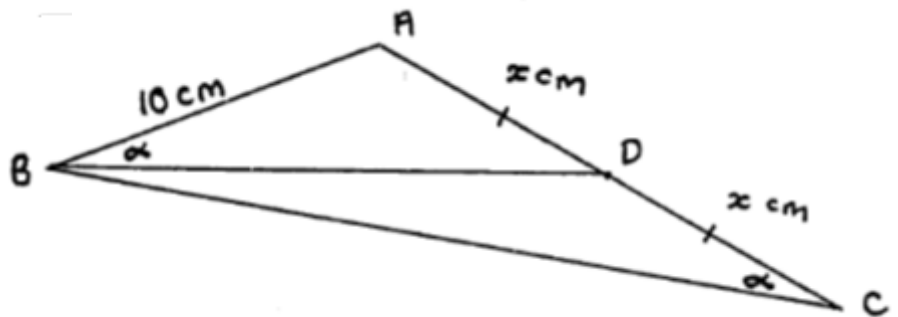
**(b)****3**

Dani and Chris used the spinner shown above to play a game. Dani spun the spinner twice and added the results of the two spins to get his score. Chris then took his turn. The player with the highest score won the game.

- i) Use a tree diagram or a sample space to show all the possible scores Dani could have achieved when he played the game.
- ii) What is the probability that Dani scored 6 in the game?
- iii) Dani's score was 6. What is the probability that Chris won the game?

**Question 13 is continued over**

(c)



In the diagram above,  $AB = 10 \text{ cm}$ ,  $AD = DC = x \text{ cm}$  and  $\angle ABD = \angle BCD$ .

4

i) Prove  $\triangle ABD \sim \triangle ACB$

ii) Hence find the exact value of  $x$ .

iv) If  $\alpha$  and  $\beta$  are the roots of the quadratic equation  $5x^2 - 3x - 2 = 0$ , find the value of:

4

i)  $\alpha + \beta$

ii)  $\alpha\beta$

iii)  $\alpha^2 + \beta^2$

iv)  $(3 - \alpha)(3 - \beta)$

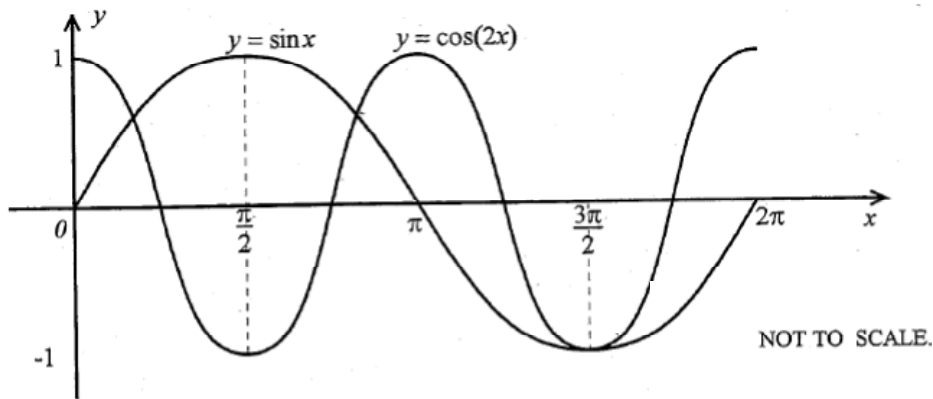
**Question14 (15 marks) Begin a NEW booklet****marks**

- (a) Rapunzel has discovered something interesting about the way her hair grows. **6**  
When she was imprisoned in her tower ( on the 1<sup>st</sup> of January ) her hair was 9 metres long. She measured her hair every year on the 1<sup>st</sup> of January, and made a note of its length. She recorded the following lengths on her cell wall for the first 4 years of her imprisonment:

9 m, 10 m, 11.05 m, 12.15 m, 13.3 m

- i) How much did Rapunzel's hair grow in the second year?  
ii) Write a series showing the amount her hair had grown in each of the first 4 years.  
iii) Using the series in part (ii), find how much Rapunzel's hair grew in the eighteenth year.  
iv) Prince Charming arrived to rescue Rapunzel after she had been in the tower for 21 years. If the tower is 40 metres high, and assuming that her hair continues to grow according to the series in (ii), will Rapunzel's hair be long enough to reach the ground ? Justify your answer by finding the length of Rapunzel's hair at this time.

(b)



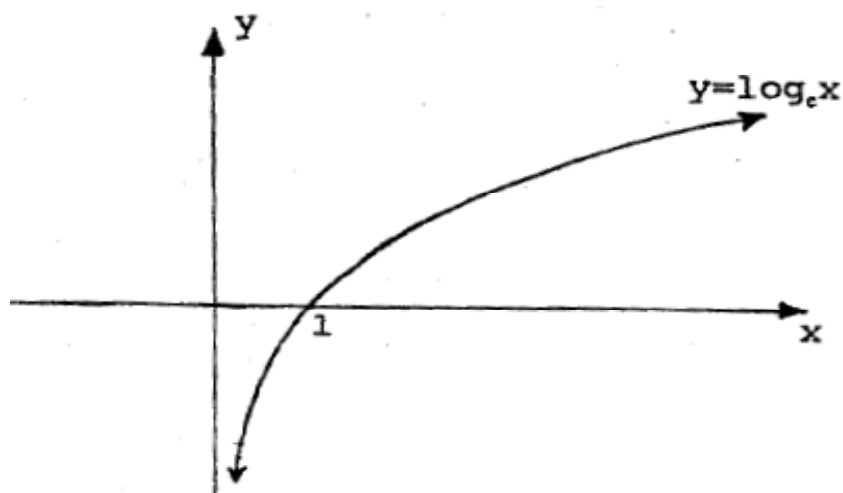
The diagram shows parts of the curves  $y = \sin x$  and  $y = \cos 2x$ .

**6**

- i) Show that the curves intersect at  $x = \frac{5\pi}{6}$  and  $x = \frac{3\pi}{2}$ .  
ii) Calculate the area between the curves from  $x = \frac{5\pi}{6}$  to  $x = \frac{3\pi}{2}$ . Leave the answer in exact form.
- c) The region enclosed by the  $y$  axis, the line  $y = x - 1$  and the line  $y = 3$  is rotated about the  $y$  axis.
- i) Sketch the region **1**  
ii) Show that the volume of the resulting solid of revolution is given by **1**  
$$V = \pi \int_{-1}^3 (y^2 + 2y + 1) dy$$
  
iii) Hence, calculate this volume correct to two decimal places. **1**

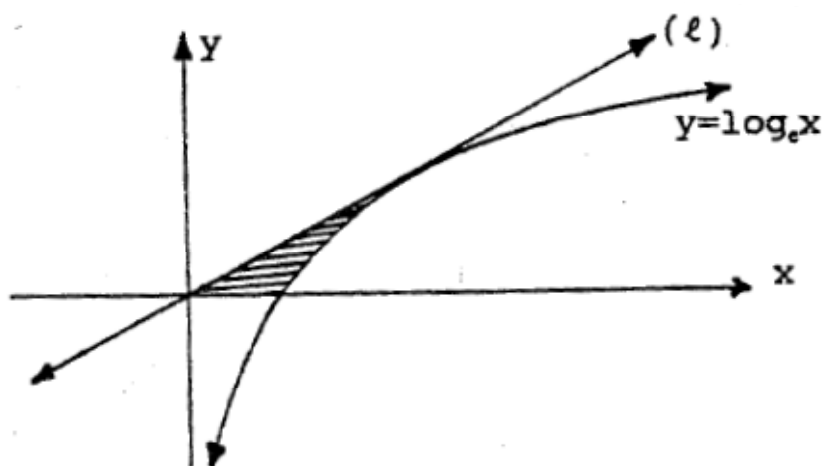
(a)

6



The diagram shows the graph of the curve  $y = \log_e x$ , not drawn to scale.

- (i) Find the equation of the tangent ( $l$ ) at  $x = e$  to the curve and show that it passes through the origin.
- (ii) Show that the area bounded by the curve, the y-axis and the lines  $y = 0$  and  $y = 1$  is  $(e - 1)$  square units.
- (iii) Using the results of part (ii) or otherwise, find the area of the shaded region between  $y = \log_e x$ , the tangent ( $l$ ) and the x-axis as shown in the diagram below.



Question 15 is continued over

- b) A river 60m wide is measured for depth every 10m across its width. 2

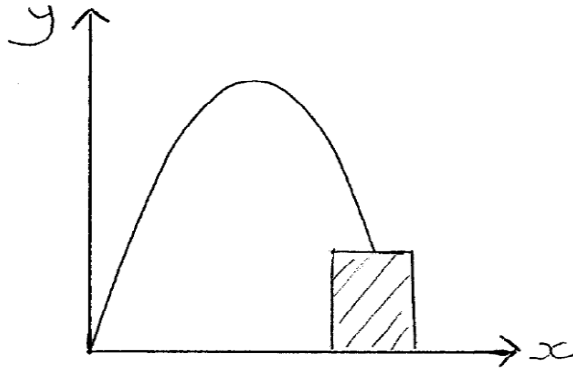
The measurements from bank to bank are given in the table. By using Simpson's Rule, estimate the cross-sectional area of the river at this point.

|                    |   |     |      |      |     |     |   |
|--------------------|---|-----|------|------|-----|-----|---|
| Measurement number | 1 | 2   | 3    | 4    | 5   | 6   | 7 |
| Depth in metres    | 0 | 5.5 | 11.0 | 13.2 | 8.5 | 4.5 | 0 |

- (c) (i) Expand  $e^{-x}(1 - e^{-x})$  1
- (ii) For the curve  $y = e^{-x} - e^{-2x}$ : 6
- (α) find where it cuts the axes.
- (β) find the co-ordinates of the stationary points and determine their nature.
- (γ) determine the values of  $x$  for which the curve is monotonic decreasing and hence, or otherwise, discuss the behaviour of the curve for large values of  $x$ .
- (δ) sketch the curve.

**Question 16 (15 marks) Begin a NEW booklet****marks**

- (a) A water spout reaches a maximum height of 40m at a point 20m away from its source on the ground. It lands on the roof of a building 25 m away. How high is the building? (A water spout traces out a parabola.)

**3**

- (b) Fabio and Su Lin worked out that they would save \$50000 in 5 years by depositing all their combined monthly salary of \$S at the beginning of each month into a savings account and withdrawing \$1600 at the end of each month for living expenses. The savings account paid interest at the rate of 3% p.a. compounding monthly.

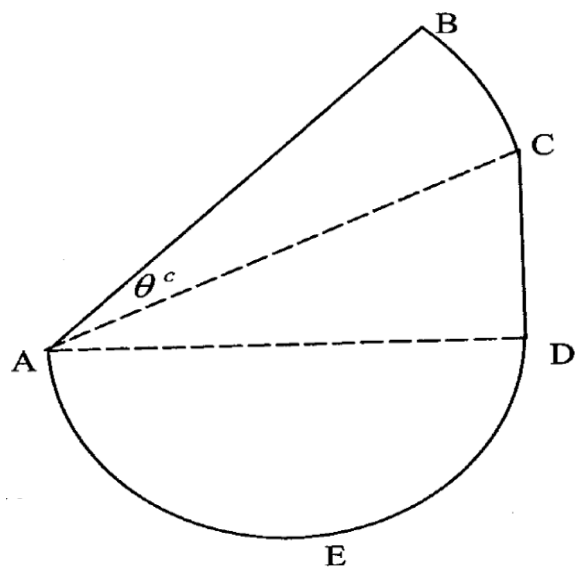
**5**

- (i) Show that at the end of the second month, the balance in their savings account, immediately after making their withdrawal of \$1600, would be given by  $[(1.0025^2 + 1.0025)S - 1600(1.0025 + 1)]$ .
- (ii) Hence write down an expression for the balance in their account at the end of the 60<sup>th</sup> month.
- (iii) Calculate their combined monthly salary.

Question 16 continued

(c)

7



ABC is a sector of centre A and radius AB.

ACD is a right-angled triangle with  $\angle ADC = 90^\circ$ .

AED is a semi-circle of radius  $2r$ .  $CD = 3r$ . The perimeter of ABCDE is 24m.

(i) Show that  $\theta = \frac{24 - 2\pi r - 8r}{5r}$

(ii) Show that the total Area =  $60r - 14r^2 - 3\pi r^2$

(iii) Show that the total area is a maximum when  $r = \frac{30}{3\pi + 14}$

(iv) Find the maximum area to the nearest  $m^2$ .

**END OF EXAMINATION**

Q1. D Q2. D Q3. B Q4. B  
 Q5. C Q6. B Q7. A Q8. C  
 Q9. C Q10. D (10 marks)

11)  $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$  students factorised this incorrectly  
 $x^3 - 27 = (x+3)(x-3)(x+3) ?$   
 $= \lim_{x \rightarrow 3} \frac{(x-3)(x^2+3x+9)}{x-3}$  many simple maths errors.  
 $= 9+9+9$   
 $= 27$  ✓

12)  $15\sqrt{2} - 3\sqrt{3} + \sqrt{18} = a\sqrt{2} - b\sqrt{3}$  well done  
 $5\sqrt{2} - 15\sqrt{3} + 3\sqrt{2} = a\sqrt{2} - b\sqrt{3}$  ✓  
 $8\sqrt{2} - 15\sqrt{3} = a\sqrt{2} - b\sqrt{3}$   
 $a=8$  and  $b=15$  ✓

13)  $f(x) = 2\sqrt{x-1} + 3$  students used  $>$   
 D:  $x \geq 1$  ✓ instead of  $\geq$   
 R:  $y \geq 3$  ✓

14)  $\int_1^3 \frac{a}{1+x} dx = \log_e 16$  students took  $\frac{1}{a}$  out of the integral instead of  $a$   
 $a [\ln(1+x)]_1^3 = \log_e 16$   
 $a (\ln 4 - \ln 2) = \ln 16$  ✓  
 $a \ln\left(\frac{4}{2}\right) = \ln 16$   
 $a = \frac{\ln 16}{\ln 2}$   
 $= \frac{4 \ln 2}{\ln 2}$   
 $a = 4$  ✓  
 students left the answer as  $\frac{\ln 16}{\ln 2}$

$2x-1 = 3x-4$

$2x-1 = 3x-4$   
 $3 = x$

check: LHS = 5  
 RHS = 5

$-2x+1 = 3x-4$   
 $5 = 5x$   
 $x = 1$

check: LHS = 1  
 RHS = -1  
 $\therefore$  not a soln

$\therefore$  the only soln is  $x=3$

15)  $x^2 - (k-3)x + k = 0$   
 for equal roots  $\Delta = 0$   
 i.e.  $b^2 - 4ac = 0$   
 $[-(k-3)]^2 - 4 \times 1 \times k = 0$  ✓  
 $k^2 - 6k + 9 - 4k = 0$   
 $k^2 - 10k + 9 = 0$   
 $(k-9)(k-1) = 0$   
 $\therefore k = 1$  or  $9$  ✓

16)  $3y - 2x + y = 180$  (co-int. L's, EF || DC)  
 $4y - 2x = 180$   
 $2y - x = 90$  ①  
 $78 + x + y = 180$  (co-int. L's, AB || CD)  
 $x + y = 102$  ②

D+②:  $3y = 192$   
 $y = 64$  ✓  
 Sub in ②:  $x + 64 = 102$   
 $x = 38$  ✓

mostly well done

students did not check their answers.  
 must check both solns

mostly well done

1 for setting up eqns

$$1) y = \sqrt{1-x^2}$$

$$y = (1-x^2)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \times -2x$$

$$\boxed{\frac{dy}{dx} = \frac{-x}{\sqrt{1-x^2}}}$$

✓  
accept  
 $\frac{dy}{dx} = -x(1-x^2)^{-1/2}$

Watch  
Chain  
rule

$$ii) y = \frac{\cos x}{x}$$

$$\frac{dy}{dx} = \frac{v \frac{dv}{dx} - u \cdot \frac{du}{dx}}{v^2}$$

$$= \frac{x \cdot -\sin x - \cos x \cdot 1}{x^2}$$

$$\boxed{\frac{dy}{dx} = \frac{-x \sin x - \cos x}{x^2}}$$

Well  
done

$$2) i) \int \frac{1}{e^{3x}} dx = \int e^{-3x} dx$$

$$= -\frac{1}{3} e^{-3x} + C$$

Some  
confusion  
 $\neq -3e^{-3x}$

$$ii) \int \left(1 + 3 \sec^2 \frac{x}{2}\right) dx$$

$$= x + 3 \tan \frac{x}{2} + C$$

$$= x + 6 \tan \frac{x}{2} + C$$

Well  
done

$$3) f'(x) = 6x^2 - 6x + 5$$

$$f(x) = \frac{6x^3}{3} - \frac{6x^2}{2} + 5x + C$$

$$f(x) = 2x^3 - 3x^2 + 5x + C$$

$$\text{at } (2, 13) \quad 13 = 16 - 12 + 10 + C$$

$$C = -1$$

$$\therefore f(x) = 2x^3 - 3x^2 + 5x - 1$$

Wide range  
of constants.

$$1) y = \ln \left( \frac{2x-1}{x+1} \right)$$

$$y = \ln(2x-1) - \ln(x+1)$$

$$y' = \frac{2}{2x-1} - \frac{1}{x+1}$$

$$\text{at } x=2: y' = \frac{2}{3} - \frac{1}{3}$$

$$= \frac{1}{3}$$

$$\Rightarrow m = \frac{1}{3}$$

$$\therefore \perp m = -3$$

$$\text{then } x=2: y = \ln 1 = 0$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = -3(x - 2)$$

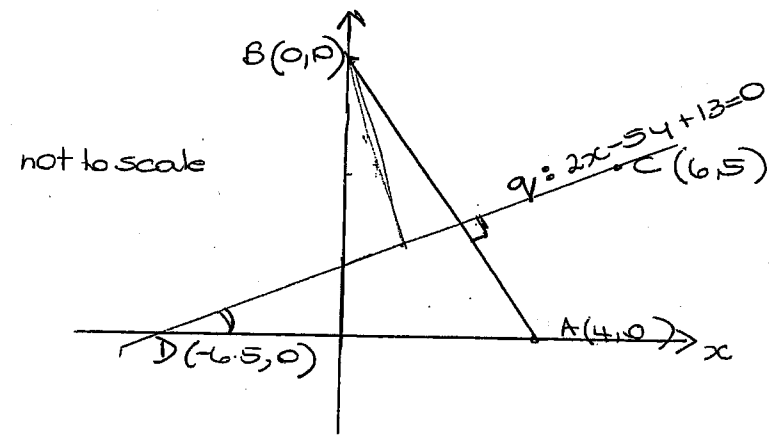
$$\boxed{y = -3x + 6}$$

or

$$\boxed{3x + y - 6 = 0}$$

Some  
log different  
problems

$\Rightarrow$



$$i) m_{AB} = \frac{10-0}{0-4}$$

$$m_{AB} = -\frac{5}{2}$$

$$ii) \perp m = \frac{2}{5}$$

$$y - y_1 = m(x - x_1)$$

$$y - 5 = \frac{2}{5}(x - 6)$$

$$5y - 25 = 2x - 12$$

$$2x - 5y + 13 = 0$$

(as req'd)

for  
working

Mostly  
Well  
done

$$a=2 \quad b=-5 \quad c=13$$

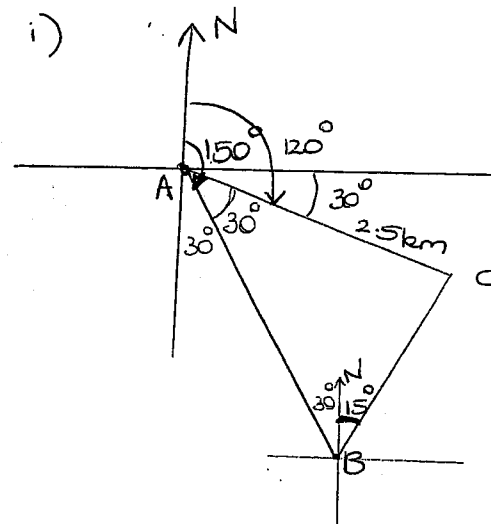
$$\begin{aligned} \text{sp. dist.} &= \left| \frac{ax+c}{\sqrt{a^2+b^2}} \right| \\ &= \left| \frac{0-50+13}{\sqrt{4+25}} \right| \\ &= \left| \frac{-37}{\sqrt{29}} \right| \\ &= \frac{37}{\sqrt{29}} \times \frac{\sqrt{29}}{\sqrt{29}} \\ &= \frac{37\sqrt{29}}{29} \text{ units.} \end{aligned}$$

Some people didn't rationalise as asked

$$\begin{aligned} \checkmark) \quad CD(9) \text{ cuts } x \text{ axis} &\Rightarrow y=0 \\ 2x &=-13 \\ x &=-6.5 \Rightarrow D(-6.5, 0) \text{ not needed.} \\ \text{now } m &= \frac{2}{9} \text{ from ii)} \\ \therefore \tan \theta &= m \\ \tan \angle ADC &= \frac{2}{9} \\ \angle ADC &= 22^\circ \text{ (nearest deg.)} \end{aligned}$$

$$\frac{21.8^\circ}{\text{OK}}$$

9) i)



$$\text{ii)} \quad \frac{AB}{\sin 105^\circ} = \frac{2.5}{\sin 45^\circ}$$

$$AB = \frac{2.5 \sin 105^\circ}{\sin 45^\circ}$$

$$= 3.415063509 \checkmark$$

$$AB = 3.42 \text{ km (2dp)}$$

$$\frac{BC}{\sin 30^\circ} = \frac{2.5}{\sin 45^\circ}$$

$$BC = \frac{2.5 \times \frac{1}{2}}{\frac{1}{\sqrt{2}}}$$

$$= 1.767766953 \checkmark$$

$$BC = 1.77 \text{ km (2dp)}$$

$$\begin{aligned} \text{iii)} \quad \text{av. speed} &= \frac{D}{T} \\ &= \frac{AB+BC}{2} \end{aligned}$$

$$= 2.591415231$$

$$= 2.6 \text{ km/h (correct to 1 dp)}$$

\* OR 2.6 km/h if using full display on calculator

\* Diagrams were very poorly done. They must be  $\frac{1}{3}$  A4 page in pencil using a ruler!

\* many students did not mark on the given info as asked. They marked 'deduced' info instead.

Errors in ii) were mainly due to incorrect angle sizes.

\* some tried to use right triangle ratios ??

\* many students included AC!

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 2 | 3 | 4 | 5 | 6 |
| 3 | 4 | 5 | 6 | 7 |
| 4 | 5 | 6 | 7 | 8 |

or can use  
a tree diagram

\* fairly done  
many students  
recorded the  
outcomes, not  
the scores as  
asked in the  
question.

generally well  
done!

\* MOST STUDENTS  
NEED TO REVISE  
SETTING OUT  
SIMILARITY  
PROOFS and  
the CORRECT TESTS!!!

(Many are confusing  
congruence tests  
with similarity  
tests)

→ reason should  
be stated for  
setting up the  
the proportion  
statement

\* generally very  
well done but  
some students  
must learn the  
main expressions  
 $x+b$ ,  $xb$ ,  $(x+b)^2$

14

(a)  $9m, 10m, 11.05m, 12.15m, 13.3m$

i)  $1.05m$  ✓

ii)  $1 + 1.05 + 1.10 + 1.15$  ✓

iii)  $T_n = a + (n-1)d$  AP:  $a=1, d=0.05$

$T_{18} = 1 + 17 \times 0.05$   
 $= 1.85$  ✓

∴ grew  $1.85m$  in the 18th year

iv)  $S_n = \frac{n}{2} [2a + (n-1)d]$

$S_{21} = \frac{21}{2} [2 + 20 \times 0.05]$  ✓

$S_{21} = 31.5$

Length of hair after 21 yrs

$= 9 + S_{21}$

$= 40.5m$  ✓

Since the tower is  $40m$  high, her  
hair will be long enough. ✓

b)  $y = \sin x$

$x = \frac{5\pi}{6}$  :  $y = \sin \frac{5\pi}{6}$

$y = \frac{1}{2}$

$y = \cos 2x$

$= \cos 2(\frac{5\pi}{6})$

$= \cos \frac{5\pi}{3}$

$= \frac{1}{2}$

∴ curves intersect at  $x = \frac{5\pi}{6}$  ✓

$x = \frac{3\pi}{2}$  :  $y = \sin \frac{3\pi}{2}$

$= -1$

$y = \cos 2(\frac{3\pi}{2})$

$= -1$

∴ curves intersect at  $x = \frac{3\pi}{2}$  ✓

ii)  $A = \int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} (\cos 2x - \sin x) dx$

$= \left[ \frac{1}{2} \sin 2x + \cos x \right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}$  ✓

$= \left( \frac{1}{2} \sin 3\pi + \cos \frac{3\pi}{2} \right) - \left( \frac{1}{2} \sin \frac{5\pi}{3} + \cos \frac{5\pi}{6} \right)$

$= (0 + 0) - \left( -\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{2} \right) + \cos \frac{5\pi}{6}$

$= \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}$  ✓

\* Most students are not reading  
the requirements of the question

ie (i) 2nd year, not 1st!

(ii) "Write a series..."

(iii) "...in the 18th year..."

\* many did not know the  
definition of a series.

\* Several students found  
 $T_{21}$  instead of  $S_{21}$ .

\* many forgot to add the  $9m$   
growth to the initial  $9m$   
length, thus forever trapping  
Rapunzel unjustly in her tower  
forever.

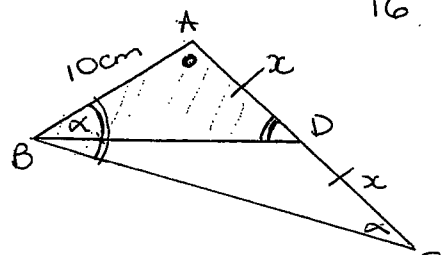
\* while many made and solved  
the equation  $\cos 2x = \sin x$ ,  
too many used an equation to  
show. Solve or show, don't  
mix the methods!

Show  
substitutions

\* despite being clearly asked  
to evaluate between  $\frac{5\pi}{6}$  and  $\frac{3\pi}{2}$ ,  
many tried to add together every  
possible area/integral (wasting  
time).

\* Many also seem unaware  
that  $\frac{5\pi}{6} < \frac{3\pi}{2}$ .

\* exact value - this doesn't mean  
leave in trig form when the angle  
gives exact trig values!!



i) In  $\Delta$ 's  $ABD$  &  $ACB$

$\angle A$  is common

$\angle ABD = \angle ACB (= \alpha)$  given

∴  $\Delta ABD \parallel \Delta ACB$  equiangular ✓

∴  $\frac{AD}{AB} = \frac{AB}{AC}$  (corres. sides of  
sim.  $\Delta$ 's are in  
same ratio)

$\frac{x}{10} = \frac{10}{2x}$  ✓

$2x^2 = 100$

$x^2 = 50$

$x = 5\sqrt{2}$  ✓ ( $x > 0$ )

ii)  $5x^2 - 3x - 2 = 0$   
 $a=5$   $b=-3$   $c=-2$

i)  $x+b = \frac{-b}{a}$   
 $x+b = \frac{3}{5}$  ✓

iii)  $x^2+b^2 = (x+b)^2 - 2xb$

$= \frac{9}{25} + \frac{4}{5}$

$= \frac{29}{25}$  ✓

ii)  $xb = \frac{c}{a}$   
 $xb = \frac{-2}{5}$  ✓

iv)  $(3-x)(3+x) = 9 - 3x + 3x + x^2$   
 $= 9 - 3(x+b) + x^2$

$= 34$  ✓

$x = y+1$   
 $x^2 = (y+1)^2$   
 $x^2 = y^2 + 2y + 1$

$V = \pi \int x^2 dy$   
 $= \pi \int (y^2 + 2y + 1) dy$   
 $= \pi \left[ \frac{y^3}{3} + \frac{2y^2}{2} + y \right]_{-1}^3$   
 $= \pi \left[ (9 + 9 + 3) - \left(-\frac{1}{3} + 1 - 1\right) \right]$   
 $= \pi \left[ 21\frac{1}{3} \right]$   
 $= \frac{64\pi}{3} \text{ u}^3$   
 $= 67.02 \text{ u}^3 \text{ (204p)}$

show expansion for mark

Sketch the region - 4.  
 you must indicate the region somehow: shading is the standard way to do this. Simply drawing all the lines does not fulfil the requirements of the question!

As the integral is already given, you must make the link between (i) as the starting point and this integral! Too many wrote  $V = \pi \int_{-1}^3 x^2 dy$   
 $= \pi \int_{-1}^3 (y^2 + 2y + 1) dy$   
 (this scored no marks)

too many stopped at  $V = \frac{64}{3}\pi$  (those that could at least get the  $\frac{64}{3}$ ) and did not fulfil the requirements of the question

$\pi$  also had a habit of mysteriously disappearing - leaving  $21\frac{1}{3}$  as a common answer!

# Question 15 (15 marks)

comments

a) i)  $y = \ln x$   
 $y' = \frac{1}{x}$

at  $x = e$ ,  $y' = \frac{1}{e}$ ,  $y = \ln e = 1$

Eqn of tangent:

$y - 1 = \frac{1}{e}(x - e)$  ← ① mark for eqn. of tangent

$ey - e = x - e$   
 $x - ey = 0$

sub (0,0) into

$x - ey = 0$

$0 - e(0) = 0$

$0 = 0$  ← ① mark for showing origin is on tangent.  
 LHS = RHS

tangent passes through the origin.

ii)  $A = \int_0^1 e^y dy$  ←  $y = \ln x$   
 $\therefore x = e^y$  ← ① mark for integral

$= \left[ e^y \right]_0^1$   
 $= e^1 - e^0$  ← ① mark for working.

$= (e - 1) \text{ units}^2$

iii)  $A = (e - 1) - A \text{ of } \Delta$

$= (e - 1) - \frac{e \times 1}{2}$  ← ① mark

$= e - 1 - \frac{e}{2}$   
 $= \left(\frac{e}{2} - 1\right) \text{ units}^2$  ← ① mark

b)  $h = 10$

|       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|
| 0     | 10    | 20    | 30    | 40    | 50    | 60    |
| 0     | 5.5   | 11    | 13.2  | 8.5   | 4.5   | 0     |
| $y_1$ | $y_2$ | $y_3$ | $y_4$ | $y_5$ | $y_6$ | $y_7$ |

$A = \frac{h}{3} \{ y_1 + y_7 + 4(y_2 + y_4 + y_6) + 2(y_3 + y_5) \}$   
 $= \frac{10}{3} \{ 0 + 0 + 4(5.5 + 13.2 + 4.5) + 2(11 + 8.5) \}$  ← ① mark  
 $= 439.3 \text{ units}^2$  ← ① mark

well answered

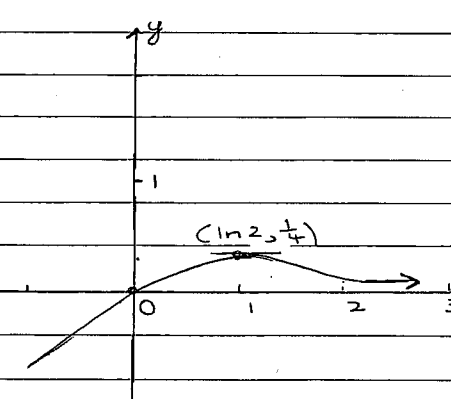
many forgot to read the q carefully and did not attempt this part.

A common mistake was to try and  $\int \ln x dx$

More CARE needed.

Many students did not recognise  $h = 10$  from q.

| 15 continued.                                                                                                                                                                                | comments                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 2) i) $e^{-x}(1-e^{-x})$<br>$= e^{-x} - e^{-2x}$                                                                                                                                             | ① for expanding correctly                                                                                   |
| ii) $y = e^{-x} - e^{-2x}$                                                                                                                                                                   |                                                                                                             |
| x) sub $x=0$ : sub $y=0$<br>$y = e^0 - e^0 = 1-1 = 0$<br>$e^{-x} \neq 0, e^{-x} = 1$ when $x=0$                                                                                              | some errors in not recognising $e^{-x} \neq 0$ and when $e^{-x} = 1$ $x=0$                                  |
| $\therefore$ cuts $x$ & $y$ axes at origin.                                                                                                                                                  | ① for origin.                                                                                               |
| 3) $\frac{dy}{dx} = -e^{-x} + 2e^{-2x}$                                                                                                                                                      |                                                                                                             |
| at stationary pts, $\frac{dy}{dx} = 0$                                                                                                                                                       |                                                                                                             |
| $-e^{-x} + 2e^{-2x} = 0$<br>$e^{-x}(2e^{-x} - 1) = 0$<br>$e^{-x} \neq 0, e^{-x} = \frac{1}{2}$<br>$\ln(\frac{1}{2}) = -x$<br>$x = -\ln(\frac{1}{2})$<br>$x = (\ln 1 - \ln 2)$<br>$x = \ln 2$ | many students had trouble solving<br>$2e^{-x} - 1 = 0$<br>$\Downarrow \Downarrow \Downarrow$<br>$x = \ln 2$ |
| If $x = \ln 2$ , $y = e^{-\ln 2} - e^{-2\ln 2}$<br>$= \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$                                                                                               |                                                                                                             |
| $\therefore$ stationary pt. is at $(\ln 2, \frac{1}{4})$<br>$(\approx 0.69, \frac{1}{4})$                                                                                                    | ① for stat. pt.                                                                                             |
| $\frac{d^2y}{dx^2} = e^{-x} - 4e^{-2x}$                                                                                                                                                      |                                                                                                             |
| at $x = \ln 2$ , $\frac{d^2y}{dx^2} < 0$                                                                                                                                                     |                                                                                                             |
| $\therefore (\ln 2, \frac{1}{4})$ is a maximum turning pt.                                                                                                                                   | ① for nature of stat. pt.                                                                                   |

| Question 15 continued.                                                                                                                                                                                                                                                                            | comments                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| c) ii) x) Since there is only one turning point and it is a maximum turning point, then curve is monotonically decreasing for $x > \ln 2$ .                                                                                                                                                       | ① for $x > \ln 2$                                                                            |
| OR<br>$\frac{dy}{dx} = -e^{-x} + 2e^{-2x} < 0$<br>i.e. $e^{-x}(2e^{-x} - 1) < 0$<br>when $2e^{-x} - 1 < 0$ as $e^{-x} > 0$<br>$\therefore e^{-x} < \frac{1}{2}$ for all real $x$<br>i.e. $\ln e^{-x} < \ln \frac{1}{2}$<br>$-x < \ln \frac{1}{2}$<br>$x > -\ln \frac{1}{2}$<br>i.e. $x > \ln 2$ . | poorly answered<br>many students found their answer to be $x < \ln 2$ instead of $x > \ln 2$ |
| as $x \rightarrow +\infty$ , $y = e^{-x}(1-e^{-x}) \rightarrow 0$<br>(because $e^{-x} \rightarrow 0$ )                                                                                                                                                                                            | ① for $y \rightarrow 0$ well answered                                                        |
| 5)                                                                                                                                                                                                                                                                                                |                                                                                              |
|                                                                                                                                                                                                               | ① for sketch<br>poorly answered and often contradicted previous working.                     |

| Question 16 (15 marks)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Comments                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) eqn of parabola is in form: $(x-h)^2 = -4a(y-k)$<br>e. $(x-20)^2 = -4a(y-40)$<br>Find 'a':<br>$(0-20)^2 = -4a(0-40)$<br>$400 = 160a$<br>$a = 2.5$<br>∴ eqn. of parabola:<br>$(x-20)^2 = -10(y-40)$<br>When $x = 25$ :<br>$(25-20)^2 = -10(y-40)$<br>$25 = -10y + 400$<br>$-375 = -10y$<br>∴ $y = 37.5$<br>∴ height of building is 37.5 metres.                                                                                                                                                                                                            | ① mark.<br>* vertex $(h, k) = (20, 40)$<br>• extension / students tried to use projectile motion formulae.<br>① for 'a' • students used $y = ax^2 + bx + c$ which turns out to be much more difficult.<br>• calculation error was common |
| 2) $r = 3\% \text{ p.a. } a = 0.25\% / \text{month}$<br>$A_n =$ balance in savings account after $n$ months, after withdrawal<br>$A_1 = S(1.0025) - 1600$ ← ① mark<br>$A_2 = (A_1 + S)1.0025 - 1600$<br>$= A_1(1.0025) + S(1.0025) - 1600$<br>$= [S(1.0025 - 1600)]1.0025 + S(1.0025) - 1600$<br>$= S(1.0025)^2 - 1600(1.0025) + S(1.0025) - 1600$<br>$= S[1.0025 + 1.0025^2] - 1600[1 + 1.0025]$<br>① mark for deriving $A_2$<br>i) $A_{60} = S[1.0025 + 1.0025^2 + \dots + 1.0025^{60}] - 1600[1 + 1.0025 + \dots + 1.0025^{59}]$<br>⇒ ① mark for $A_{60}$ |                                                                                                                                                                                                                                          |

| Question 16 ct'd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Comments |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| b) iii) $A_{60} = 50000$<br>$50000 = S(1.0025 + \dots + 1.0025^{60}) - 1600(1 + 1.0025 + \dots + 1.0025^{59})$<br>$50000 = S \left[ \frac{1.0025(1.0025^{60} - 1)}{0.0025} \right] - 1600 \left[ \frac{1(1.0025^{60} - 1)}{0.0025} \right]$ ← ① mark<br>$50000 = S \left[ 401(1.0025^{60} - 1) \right] - 1600 \left[ 400(1.0025^{60} - 1) \right]$<br>$S = \frac{50000 + 640000[1.0025^{60} - 1]}{401(1.0025^{60} - 1)}$<br>$S = \dots 2367.56$ ← ① mark<br>• students made the second geometric sum with 59 terms. |          |
| students made $S_{60} = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |

# Question 16 of 1d

comments

c) i) In right  $\triangle ADC$ :

$AC = 5r$  (Pythagoras' Thm)

$\therefore AB = 5r$

$AB + BC + CD + DEA = 24$  (given)

$\therefore 5r + (5r)\theta + 3r + \frac{1}{2} \times \pi \times 4r = 24 \leftarrow \text{1 mark}$

$8r + 5r\theta + 2\pi r = 24$

$5r\theta = 24 - 2\pi r - 8r$

$\theta = \frac{24 - 2\pi r - 8r}{5r}$

$5r$

students did not show working

many wrote a "vague" equation and then just wrote the answer.

1 mark for working

As above

ii)  $A = A$  of sector  $ABC + A$  of  $\triangle ADC + A$  of semicircle

$= \frac{1}{2}(5r)^2\theta + \frac{4r \times 3r}{2} + \frac{1}{2} \times \pi \times (2r)^2 \leftarrow \text{1 mark}$

$= \frac{5r^2}{2} \left( \frac{24 - 2\pi r - 8r}{5r} \right) + 6r^2 + 2\pi r^2$

$= \frac{5r}{2} (24 - 2\pi r - 8r) + 6r^2 + 2\pi r^2$

$= 60r - 5\pi r^2 - 20r^2 + 6r^2 + 2\pi r^2$

$A = 60r - 14r^2 - 3\pi r^2$

1 mark for working

iii)  $A$  is maximum when  $A' = 0$  and  $A'' < 0$

$A' = 60 - 28r - 6\pi r$

$A' = 0$  when

$60 - 28r - 6\pi r = 0$

$60 = 6\pi r + 28r$

$60 = r(6\pi + 28)$

$r = \frac{60}{6\pi + 28} \div 2$

$r = \frac{30}{3\pi + 14}$

1 mark for finding  $r$

mostly well done

students did not test their answers.

$A'' = -28 - 6\pi < 0$

$\therefore A$  is maximum when  $r = \frac{30}{3\pi + 14}$

1 mark for testing for maximum

## 2.16 continued

Comments

iv)  $A \doteq 60(1.28) - 14(1.28)^2 -$

$3 \times \pi \times (1.28)^2$

$r = \frac{30}{3\pi + 14}$

$3\pi + 14$

$= 38 \text{ m}^2$  (nearest  $\text{m}^2$ )

$\doteq 1.28$

$\leftarrow \text{1 mark for answer}$

Calculator error occurred regularly